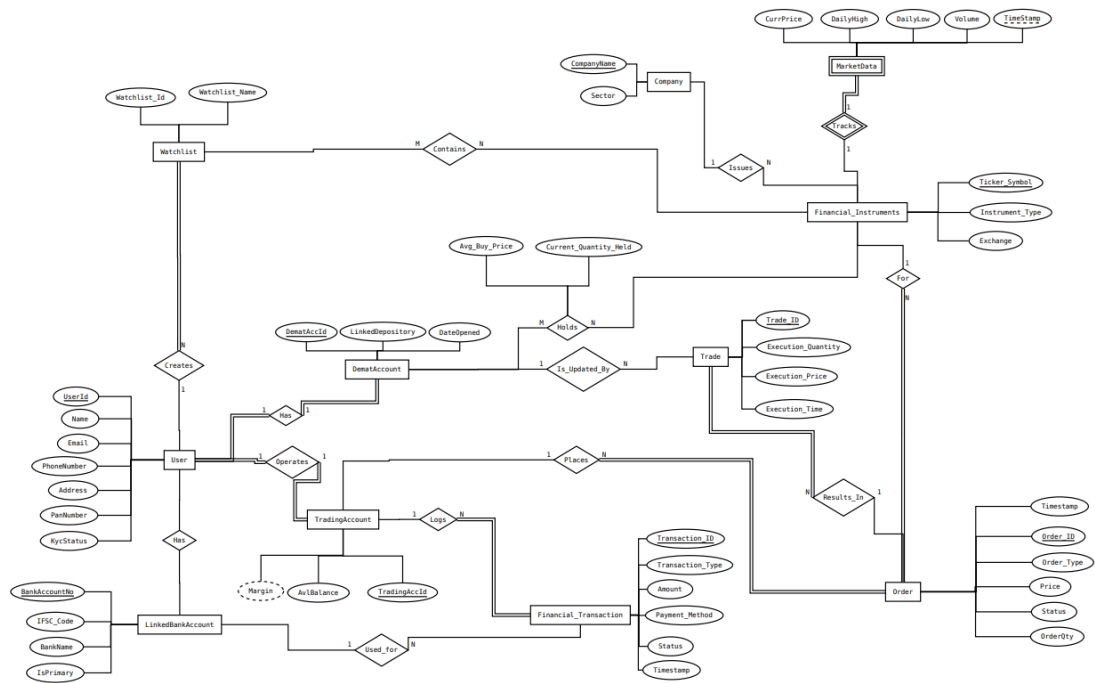
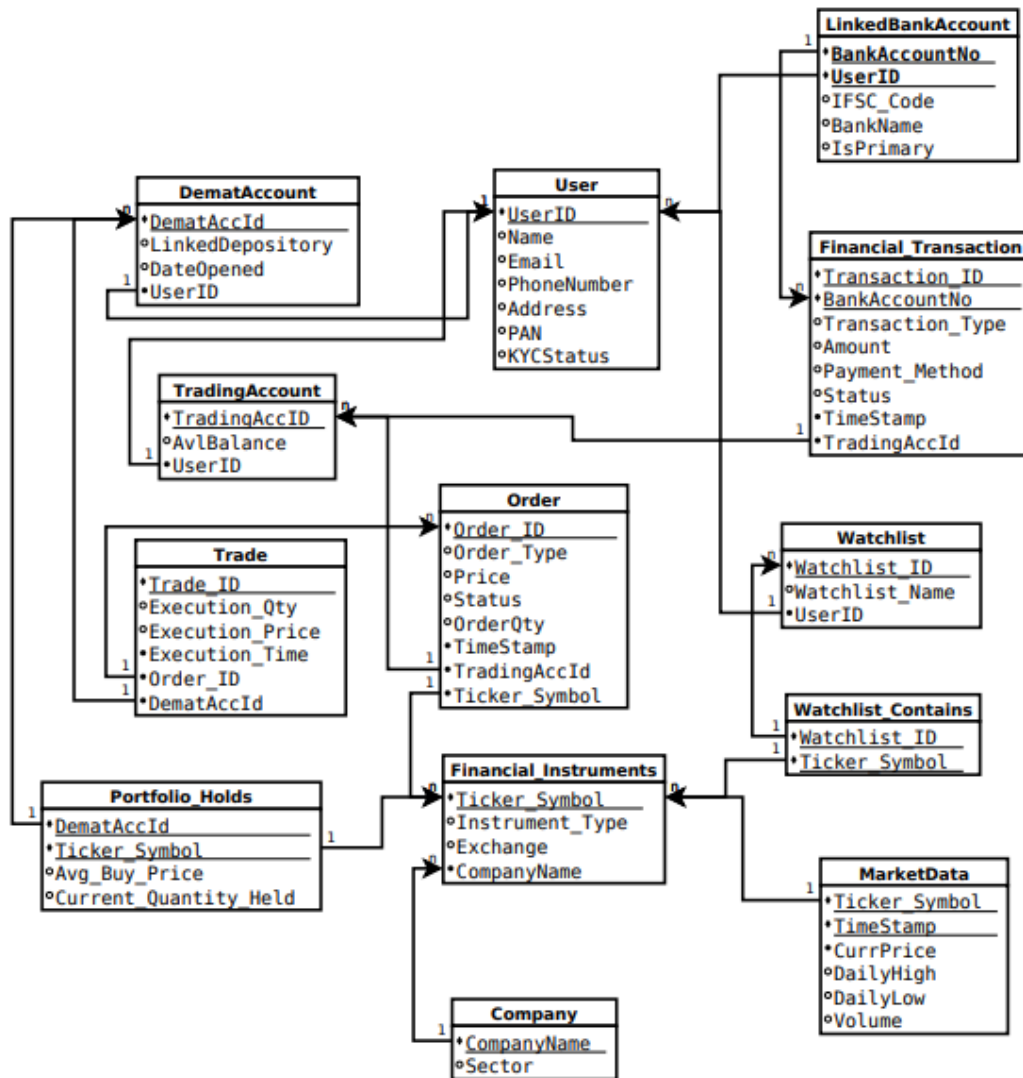


# ER



# Relational Schema



## Proof that relations are in BCNF

- To identify whether a database is in BCNF, we first determine all the functional dependencies present in each relation.
- A relation is in Boyce-Codd Normal Form (BCNF) if, for every non-trivial functional dependency  $X \rightarrow Y$ , the determinant  $X$  is a superkey. After identifying primary and candidate keys, each dependency is checked against this condition.

- For example, in the given schema, in the **AppUser** relation we have  $\text{UserId} \rightarrow \text{Name}, \text{Email}, \text{PhoneNumber}, \text{etc.}$ , where **UserId** is the primary key, so it satisfies BCNF.
- Additionally, dependencies like  $\text{Email} \rightarrow \text{UserId}$  and  $\text{PhoneNumber} \rightarrow \text{UserId}$  also hold, but since **Email** and **PhoneNumber** are defined as **UNIQUE**, they act as candidate keys, and thus still satisfy BCNF. Similarly, in the **MarketData** relation,  $(\text{Ticker\_Symbol}, \text{TimeStamp}) \rightarrow \text{DailyHigh}, \text{DailyLow}, \text{Volume}$ , and the determinant is the composite primary key, so it also satisfies BCNF.
- Since in all relations the determinants are superkeys or candidate keys, **the schema is already in BCNF**. If any dependency had a determinant that was not a superkey, we would decompose the relation into smaller relations to achieve BCNF while ensuring a lossless join.